Employee Laptop Request Management System

Project Design Phase

Project Name	Employee Laptop Request Management System
Project Type	Individual Project
Document Date	12 September 2026
Platform	ServiceNow

1. Proposed Solution

The proposed solution is a ServiceNow-based application that centralizes laptop requests and automates the approval lifecycle. Employees use a Record Producer to submit a request. The custom request table stores the information. A Business Rule checks for an annual duplicate, while Flow Designer manages approval, status updates and notifications.

2. Problem–Solution Fit

Problem	Design response	Expected value
Requests arrive through scattered emails or verbal communication.	Use one Record Producer and one request table.	Consistent request capture.
Duplicate requests may be submitted.	Use a before-insert annual duplicate check.	Improved policy enforcement.
Approval status is difficult to track.	Use Flow Designer and approval records.	Traceable manager decision.
Employees lack clear outcome updates.	Send approval/outcome notifications.	Better communication.
Access is not consistently controlled.	Use it_employee / it_manager roles and ACLs.	Role-aligned access.

3. System Architecture

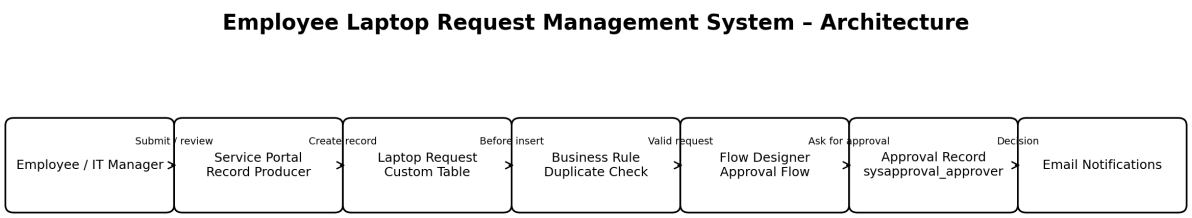


Figure 1. Logical ServiceNow architecture.

4. Request Lifecycle and Workflow

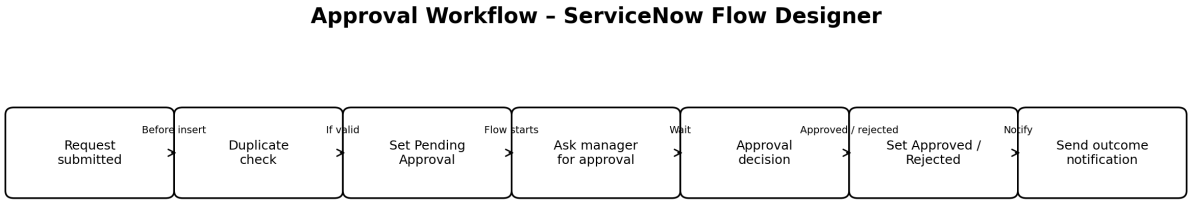


Figure 2. Manager approval workflow.

5. Logical Data Relationship

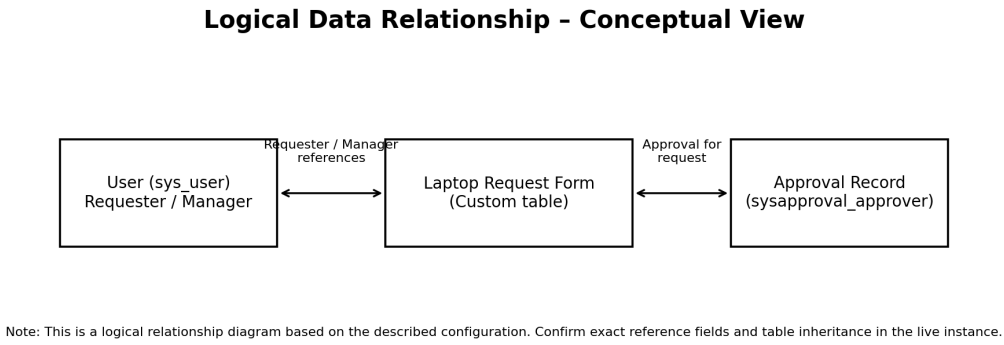


Figure 3. Conceptual data relationship between user, request and approval.

6. Main Data Fields

Field	Purpose / design note
Number	Auto-generated request identifier; should be hidden from employee form if required.
Requester	Reference to the employee/user submitting the request.
Employee Email	Contact email for notifications; populate from requester where configured.
Request Date	Date/time associated with request creation.
Short Description	Brief explanation of the laptop requirement.
Manager	Reference to the manager responsible for approval.
Type of Laptop / Request	Basic or Advance.
Reason	Software Development, Personal Use or Other.
Status	Pending Approval, Approved or Rejected.

7. Security Design

The design defines an it_employee role for permitted create/read/write operations and an it_manager role for viewing requests and acting as the approval stakeholder. Table and field ACLs must be checked in the live instance. Manager approval is performed through the approval record/Flow Designer rather than by granting broad request-record write access.

8. Design Assumptions and Constraints

The diagrams in this document are logical design illustrations derived from the supplied project documentation. They are not screenshots of a live ServiceNow instance. Exact table names, reference fields, ACL conditions and flow actions must be confirmed against the configured instance.

> Note: The conceptual data relationship diagram must not be treated as a physical database schema or an export of ServiceNow dictionary metadata.